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Level 3 — 25 marks/question

L3-Q1. count (H)

c

CODE

```
#include <stdio.h>
#include <ctype.h>
void countUpperLower(char str[],int upper,int lower){
    upper=0;
    lower=0;
    for (int i=0;str[i] !='\0';i++)
    {
        if(isupper(str[i]))
            upper++;
        else if(islower(str[i]))
            lower++;
    }
}
int main()
{
    char str[100];
    int upper,lower;
    fgets(str,sizeof(str),stdin);
    countUpperLower(str,&upper,&lower);
    printf("uppercase letters=%d\n",upper);
    printf("lowercase letters=%d\n",lower);
    return 0;
}
```

INPUT

Hello

OUTPUT

uppercase letters=268501009
lowercase letters=0

C

CODE

```
#include <stdio.h>
int isidentitymatrix(int *matrix,int n)
{
    int i,j;
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            if(i==j)
            {
                if(*(matrix + i*n + j )!=1)
                    return 0;
            }
            else
            {
                if(*(matrix + i*n + j)!=0)
                    return 0;
            }
        }
    }
    return 1;
}

int main()
{
    int matrix[3][3],i,j;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%d",&matrix[i][j]);
        }
    }
    printf("identity matrix");
    return 0;
}
```

INPUT

1 0 0
0 1 0
0 0 1

OUTPUT

identity matrix